

## Answers — items 1 and 2

The key to the two items we did **in the room**. Items 3 and 4 are **PS3**; their solutions post to bCourses after the deadline, not here. The algebra was never the hard part — the prose under each ► is.

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### 1 • Simple repression — why there is no $f$

$$F_{\text{reg}} = \frac{1}{1 + [R]/K_m}$$

► **Why there is no  $f$ .**  $f$  is the price of a **contact**: it is  $e^{-\varepsilon_{ap}/k_B T}$ , the extra Boltzmann weight of a state in which two proteins are touching each other on the DNA. In this architecture there is no such state. LacI and RNAP cannot be bound at the same time, so no state exists in which they could interact, so there is no interaction energy to charge.

The repressor works by **deleting a state from the sum**, not by changing the rate of one. That is the mechanical difference between repression-by-occlusion and activation, and it is why the two have different functional forms rather than the same form with a sign flipped.

*Worth noticing:* a repressor that binds **next to** rather than **over** the polymerase site is a different problem — that state does exist, and it carries an  $\omega$ -like factor. Not every repressor occludes.

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## 2 · Cooperative activation at $\lambda P_{RM}$

| STATE OF THE OPERATORS           | WEIGHT      | FIRES AT ( $\times \alpha$ ) |
|----------------------------------|-------------|------------------------------|
| no cl bound                      | 1           | 1                            |
| cl <sub>2</sub> on $O_{R2}$ only | $a$         | $f$                          |
| cl <sub>2</sub> on $O_{R1}$ only | $h$         | 1                            |
| both                             | $\omega ah$ | $f$                          |

Every row here has RNAP bound — the polymerase weight has already been divided out, which is why the unbound-cl row fires at 1 rather than at 0.

► **Where  $\omega$  goes.** On the **doubly-occupied state only**.  $\omega$  is an interaction between two bound dimers, so it can only appear in a state where there are two bound dimers. It is not a property of a site, and it does not multiply  $a$  or  $h$  on their own. Putting  $\omega$  on a single-occupancy state is the most common error on this item.

► **Why the  $O_{R1}$ -only state still fires at 1, not  $f$ .**  $O_{R1}$  is not in contact with polymerase. Its dimer recruits the *other* dimer; it does nothing to transcription directly. The enhancement belongs to the contact, and the contact belongs to  $O_{R2}$ .

$$F_{\text{reg}} = \frac{1 + fa + h + f\omega ah}{1 + a + h + \omega ah}$$

### The two checks

►  $\omega = 1$ , and separately  $h = 0$ . Set  $\omega = 1$  and the expression factorises:

$$F_{\text{reg}} = \frac{(1 + fa)(1 + h)}{(1 + a)(1 + h)} = \frac{1 + fa}{1 + a}$$

Set  $h = 0$  instead and you get **the same thing**, which is the single-site result from the activation table — sensitivity  $s = (\sqrt{f} - 1)/(\sqrt{f} + 1) = 0.54$  at  $f = 11$ .

► **Why two different edits give the same expression.** Because *a helper that does not interact is indistinguishable from no helper at all*. The two edits are different in the laboratory — in one, cl is sitting on  $O_{R1}$  and doing nothing; in the other,  $O_{R1}$  is not there — and the promoter cannot tell them apart. The helper occupancy cancels **completely**: not a weak effect, exactly zero.

So  $\omega$  is not a bonus on top of a second site. It is the entire mechanism by which a second site matters, and that is the design statement: **if you add an operator and get no cooperativity, you have added nothing**. A student who reports only "both give  $(1 + fa)/(1 + a)$ " has done the algebra and missed the item.

►  $[cI_2] \rightarrow \infty$ . Both  $a$  and  $h$  grow without bound — they are the same concentration over two different dissociation constants — and the  $ah$  terms dominate top and bottom:

$$F_{\text{reg}} \longrightarrow \frac{f\omega ah}{\omega ah} = f$$

$\omega$  cancels. So does  $K_{R1}$ . **The helper site cannot raise the ceiling**, and no amount of cooperativity will do it either — at saturation every dimer that can be bound is bound, and what is left is the one contact that fires the promoter. Cooperativity buys sharpness, not amplitude.

### Where $\omega$ comes from, and why two papers give different answers

The slide showed  $s = 0.93$  with the helper against **0.54** without, from Bintu et al.'s Figure 2, computed with  $f \approx 11$  and  $\omega \approx 100$ .

Ackers, Johnson & Shea measured that interaction in 1982:  $\Delta G_{12} = -1.99 \pm 0.06$  kcal/mol at 37 °C, which is  $\omega = 25$ , and their Table 3 is on the slide. So where does 100 come from? **The same laboratory, at the same temperature, ten years later.** Koblan & Ackers, *Biochemistry* 31:57 (1992), Table II, 37 °C:  $\Delta G_{12} = -2.7 \pm 0.3$  kcal/mol, which is  $\omega = 80$  with a range of 49 to 130. Bintu's 100 sits inside that. This is one group revising its own number with a better method — quantitative footprint titration analysed simultaneously across the wild-type operator and three reduced-valency mutants.

Three things are worth taking from that, and none of them is "scientists disagree."

**The 1982 value is outside the 1992 interval.**  $-1.99$  is not in  $[-3.0, -2.4]$ . The earlier number was displaced, not merely imprecise.

**The later error bar is five times wider.**  $\pm 0.06$  became  $\pm 0.3$ , because the 1992 confidence intervals include systematic differences between separate experiments and between operator templates, not only the imprecision of one fit. A tighter error bar is not automatically a better measurement. It is often a narrower definition of "error", and you cannot tell which from the number alone.

**Cooperativity is temperature-invariant; site affinity is not.** Across 5–37 °C the 1992 paper finds  $\Delta G_{12}$  between  $-1.9$  and  $-3.0$  with no interpretable trend, while the intrinsic affinities move steadily and *differentially* —  $\Delta G_1$  goes from  $-12.5$  to  $-13.5$  kcal/mol as the temperature falls. So the quantity that changes with temperature is the **discrimination between the two sites**:  $K_{R2}/K_{R1}$  is 26 at 37 °C and 92 at 5 °C. Bintu's "realistic" ratio of 25 is the 37 °C row.

### And now the result worth the whole detour

Put each temperature's *measured* pair into the expression you derived and ask what happens to the sharpness:

| T (°C) | $\omega$ | $K_{R2}/K_{R1}$ | $s$  |
|--------|----------|-----------------|------|
| 37     | 80       | 26              | 0.92 |
| 30     | 75       | 20              | 0.93 |
| 20     | 172      | 73              | 0.91 |
| 10     | 102      | 85              | 0.86 |
| 5      | 31       | 92              | 0.76 |

Both inputs swing by a factor of three to five. **The sharpness barely moves** until the cell is colder than anything  $\lambda$  normally sees. The design is robust to precisely the parameters that are worst measured — and

you could not have known that from staring at the error bars, only from pushing them through the model. That is the argument for deriving a regulation function rather than quoting one. A formula you built yourself is a formula you can feed a bad parameter and find out whether it matters.